**EX. No. 01** **WELCOME MESSAGE**

## AIM

To develop an android program for displaying the welcome message.

## PROCEDURE

1. Start an android studio application.
2. Select File in the menu bar.
3. Choose your project as **Empty View Activity** and click the Next button.
4. Name your project (The application name begins with an uppercase letter), select language as java, select your minimum API level as (Android 8.0 Oreo) then click Finish button.
5. For designing, type xml coding in **activity_main.xml**.

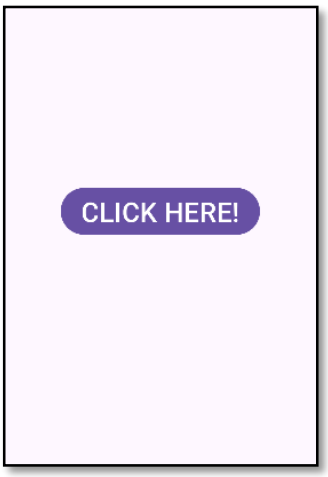
   Components used for designing

   - Button
6. Type java coding in **MainActivity.java**.
7. To run the 'app' press play button in android studio or shift + F10.
8. Select Deployment project as your Emulator
9. In the connected device, select your Emulator and click the ok button.
10. Open your installed app in Emulator.

## PROGRAM

**activity_main.xml**

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout
   xmlns:android="http://schemas.android.com/apk/res/android"
   xmlns:tools="http://schemas.android.com/tools"
   android:layout_width="match_parent"
   android:layout_height="match_parent"
   android:orientation="vertical"
   android:gravity="center"
   tools:context=".MainActivity">
<Button
   android:layout_width="wrap_content"
   android:layout_height="wrap_content"
   android:text="Click Here!"
   android:id="@+id/button"/>
</LinearLayout>
```




**MainActivity.java**

```
package com.example.toastmessage;
import androidx.appcompat.app.AppCompatActivity;
import android.os.Bundle;
import android.view.View;
import android.widget.Button;
```

```
import android.widget.Toast;

public class MainActivity extends AppCompatActivity
{
Button button;

    @Override
    protected void onCreate(Bundle savedInstanceState)
{
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
        button=findViewById(R.id.button);
        button.setOnClickListener(new View.OnClickListener()
       {
            @Override
            public void onClick(View v)
            {
                Toast.makeText(MainActivity.this, "Welcome to Android Lab",
Toast.LENGTH_LONG).show();
            }
        });

    }
}
```

**OUTPUT**

**EX. No. 02** **DISPLAYING USER INFORMATION**

## AIM

To develop an android app for displaying a form to get user information.

## PROCEDURE

1. Start an android studio application.
2. Select File in the menu bar.
3. Choose your project as **Empty View Activity** and click the Next button.
4. Name your project (The application name begins with an uppercase letter), select language as java, select your minimum API level as (Android 8.0 Oreo) then click Finish button.
5. For designing, type xml coding in **activity_main.xml**.

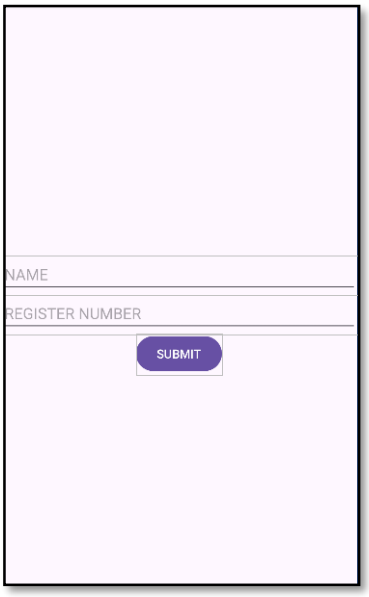
   Components used for designing

   - Button
   - EditText
6. Type java coding in **MainActivity.java**.
7. To run the 'app' press play button in android studio or shift + F10.
8. Select Deployment project as your Emulator
9. In the connected device, select your Emulator and click the ok button.
10. Open your installed app in Emulator.

## PROGRAM

**activity_main.xml**

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
   xmlns:app="http://schemas.android.com/apk/res-auto"
   xmlns:tools="http://schemas.android.com/tools"
   android:id="@+id/main"
   android:layout_width="match_parent"
   android:layout_height="match_parent"
   android:orientation="vertical"
   android:gravity="center"
   tools:context=".MainActivity">

   <EditText
      android:layout_width="match_parent"
      android:layout_height="wrap_content"
      android:id="@+id/ed1"
      android:hint="NAME">
   </EditText>

   <EditText
      android:layout_width="match_parent"
      android:layout_height="wrap_content"
      android:id="@+id/ed2"
```



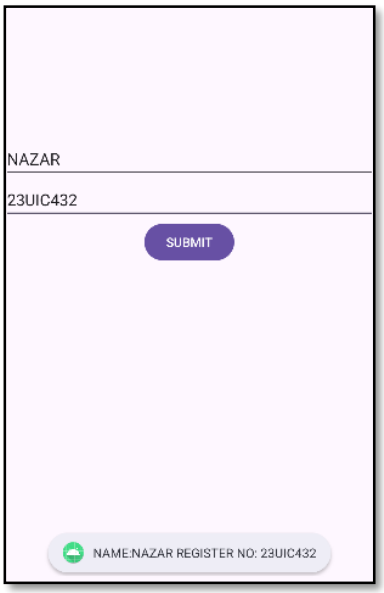
```
        android:hint="REGISTER NUMBER">
    </EditText>

    <Button
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:id="@+id/b1"
        android:text="SUBMIT">
    </Button>

</LinearLayout>
```

**MainActivity.java**

```
package com.example.userinfo;

import android.os.Bundle;
import android.view.View;
import android.widget.Button;
import android.widget.EditText;
import android.widget.TextView;
import android.widget.Toast;

import androidx.activity.EdgeToEdge;
import androidx.appcompat.app.AppCompatActivity;
import androidx.core.graphics.Insets;
import androidx.core.view.ViewCompat;
import androidx.core.view.WindowInsetsCompat;

public class MainActivity extends AppCompatActivity {
    EditText ed1,ed2;
    Button b1;


    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        EdgeToEdge.enable(this);
        setContentView(R.layout.activity_main);
        b1=findViewById(R.id.b1);
        ed1=findViewById(R.id.ed1);
        ed2=findViewById(R.id.ed2);
        ViewCompat.setOnApplyWindowInsetsListener(findViewById(R.id.main), (v, insets) -> {
            Insets systemBars = insets.getInsets(WindowInsetsCompat.Type.systemBars());
            v.setPadding(systemBars.left, systemBars.top, systemBars.right, systemBars.bottom);
            return insets;
        });
        b1.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View v) {
                String name=ed1.getText().toString();
                String regno=ed2.getText().toString();
```

```
            Toast.makeText(MainActivity.this, "NAME:"+name+ " REGISTER NO: "+regno,
Toast.LENGTH_SHORT).show();
        }
    });
  }
}
```

**OUTPUT**



**EX. No. 03** **LOGIN CREDENTIALS**

## AIM

To develop an android app for validating the login credentials and display the Welcome message using new activity for a valid user

## PROCEDURE

1. Start an android studio application.
2. Select File in the menu bar.
3. Choose your project as **Empty View Activity** and click the Next button.
4. Name your project (The application name begins with an uppercase letter), select language as java, select your minimum API level as (Android 8.0 Oreo) then click Finish button.
5. For designing, type xml coding in **activity_main.xml**.

   Components used for designing

   - Button
   - EditText
6. Type java coding in **MainActivity.java**.
7. For displaying welcome message in new activity,

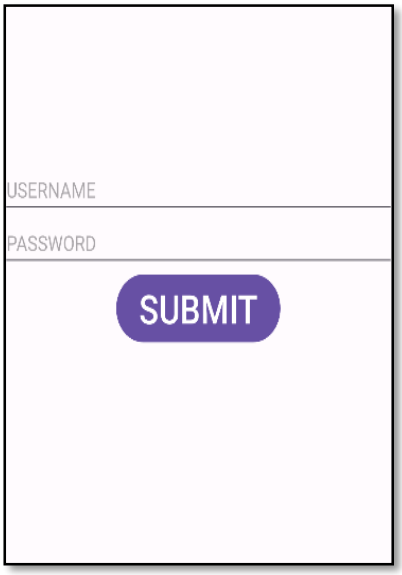
   a. Select File in menu bar
   b. Click New and select Activity
   c. Choose Empty View Activity
8. Change the activity name as Loginwelcome
9. Design the xml page (activity_loginwelcome) and type the java coding in Loginwelcome.java
10. To run the 'app' press play button in android studio or shift + F10.
11. Select Deployment project as your Emulator
12. In the connected device, select your Emulator and click the ok button.
13. Open your installed app in Emulator.

## PROGRAM

**activity_main.xml**

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:id="@+id/main"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:gravity="center"
    tools:context=".MainActivity">
```

```
<TextView
        android:id="@+id/tv1"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content" />

    <EditText
        android:id="@+id/ed1"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:ems="10"
        android:hint="USERNAME"
        android:inputType="text" />

    <EditText
        android:id="@+id/ed2"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:ems="10"
        android:hint="PASSWORD"
        android:inputType="text|textPassword" />

    <Button
        android:id="@+id/b1"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="SUBMIT"
        android:textSize="34sp" />

</LinearLayout>
```




**MainActivity.java**

```
package com.example.welcomemsg;

import android.content.Intent;
import android.os.Bundle;
import android.view.View;
import android.widget.Button;
import android.widget.EditText;
import android.widget.TextView;
import android.widget.Toast;

import androidx.activity.EdgeToEdge;
import androidx.appcompat.app.AppCompatActivity;
import androidx.core.graphics.Insets;
import androidx.core.view.ViewCompat;
import androidx.core.view.WindowInsetsCompat;

public class MainActivity extends AppCompatActivity {
    EditText usr,pass;
    Button butt;
    TextView result;
```

```
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        EdgeToEdge.enable(this);
        setContentView(R.layout.activity_main);
        usr=findViewById(R.id.ed1);
        pass=findViewById(R.id.ed2);
        butt=findViewById(R.id.b1);
        result=findViewById(R.id.tv1);
        ViewCompat.setOnApplyWindowInsetsListener(findViewById(R.id.main), (v, insets) -> {
            Insets systemBars = insets.getInsets(WindowInsetsCompat.Type.systemBars());
            v.setPadding(systemBars.left, systemBars.top, systemBars.right, systemBars.bottom);
            return insets;
        });
        butt.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View v) {
                String user=usr.getText().toString();
                String pwd=pass.getText().toString();
                result.setText("Username:"+user+" Password:"+pwd);
                if (user.equals("admin")&&pwd.equals("jmc"))
                {
                    Intent i =new Intent(MainActivity.this, Loginwelcome.class);
                    startActivity(i);
                }
                else
                {
                    Toast.makeText(MainActivity.this, "Invalid Creditionals",
Toast.LENGTH_SHORT).show();
                }
            }
        });
    }
}
```

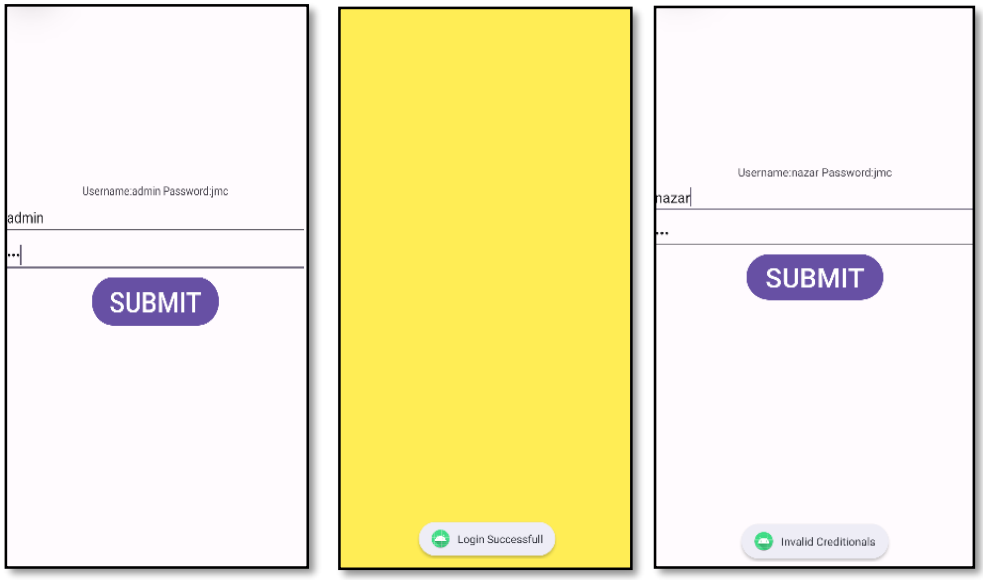
**activity_loginwelcome.xml**

```
<?xml version="1.0" encoding="utf-8"?>
<androidx.constraintlayout.widget.ConstraintLayout
xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:id="@+id/main"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:background="#DDFFEB3B"
    tools:context=".Loginwelcome">

</androidx.constraintlayout.widget.ConstraintLayout>
```

**MainActivity.java**

```
package com.example.welcomemsg;

import android.os.Bundle;
import android.widget.Toast;

import androidx.activity.EdgeToEdge;
import androidx.appcompat.app.AppCompatActivity;
import androidx.core.graphics.Insets;
import androidx.core.view.ViewCompat;
import androidx.core.view.WindowInsetsCompat;

public class Loginwelcome extends AppCompatActivity {

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        EdgeToEdge.enable(this);
        setContentView(R.layout.activity_loginwelcome);
        ViewCompat.setOnApplyWindowInsetsListener(findViewById(R.id.main), (v, insets) -> {
            Insets systemBars = insets.getInsets(WindowInsetsCompat.Type.systemBars());
            v.setPadding(systemBars.left, systemBars.top, systemBars.right, systemBars.bottom);
            return insets;
        });
        Toast.makeText(this, "Login Successfull", Toast.LENGTH_SHORT).show();
    }
}
```

**OUTPUT**



**EX. No. 04** **SIMPLE CALCULATOR**

## AIM

To develop an android app to design a calculator.

## PROCEDURE

1. Start an android studio application.
2. Select File in the menu bar.
3. Choose your project as **Empty View Activity** and click the Next button.
4. Name your project (The application name begins with an uppercase letter), select language as java, select your minimum API level as (Android 8.0 Oreo) then click Finish button.
5. For designing, type xml coding in **activity_main.xml**.

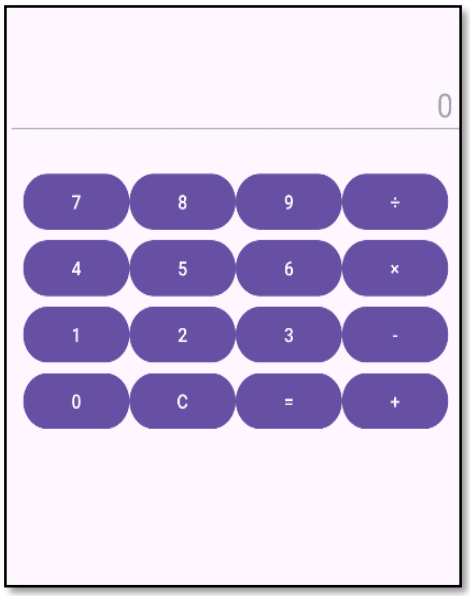
   Components used for designing

   - GridLayout
   - Button
   - EditText
6. Type java coding in **MainActivity.java**.
7. To run the 'app' press play button in android studio or shift + F10.
8. Select Deployment project as your Emulator
9. In the connected device, select your Emulator and click the ok button.
10. Open your installed app in Emulator.

## PROGRAM

**activity_main.xml**

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:id="@+id/main"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:gravity="center"
    android:padding="16dp"
    tools:context=".MainActivity">

    <EditText
        android:id="@+id/etDisplay"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:textSize="24sp"
        android:gravity="end"
        android:enabled="false"
        android:padding="10dp"
        android:hint="0" />
```



```
    <GridLayout
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:columnCount="4"
        android:layout_marginTop="20dp">

        <!-- Row 1 -->
        <Button android:text="7" android:onClick="onNumberClick"/>
        <Button android:text="8" android:onClick="onNumberClick"/>
        <Button android:text="9" android:onClick="onNumberClick"/>
        <Button android:text="÷" android:onClick="onOperatorClick"/>



        <!-- Row 2 -->
        <Button android:text="4" android:onClick="onNumberClick"/>
        <Button android:text="5" android:onClick="onNumberClick"/>
        <Button android:text="6" android:onClick="onNumberClick"/>
        <Button android:text="×" android:onClick="onOperatorClick"/>

        <!-- Row 3 -->
        <Button android:text="1" android:onClick="onNumberClick"/>
        <Button android:text="2" android:onClick="onNumberClick"/>
        <Button android:text="3" android:onClick="onNumberClick"/>
        <Button android:text="-" android:onClick="onOperatorClick"/>

        <!-- Row 4 -->
        <Button android:text="0" android:onClick="onNumberClick"/>
        <Button android:text="C" android:onClick="onClearClick"/>
        <Button android:text="=" android:onClick="onEqualClick"/>
        <Button android:text="+" android:onClick="onOperatorClick"/>

    </GridLayout>

</LinearLayout>
```

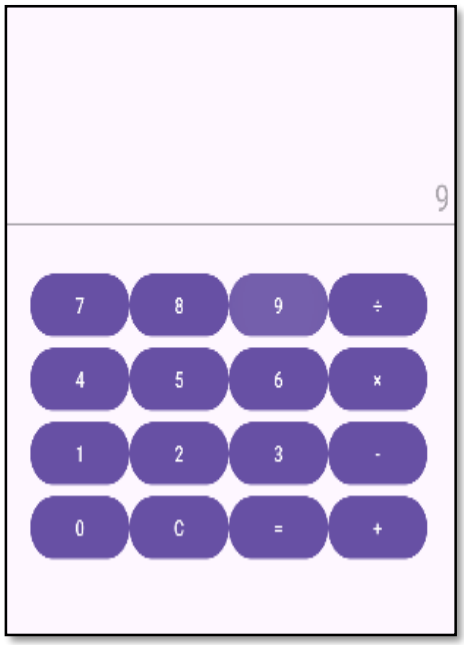
**MainActivity.java**

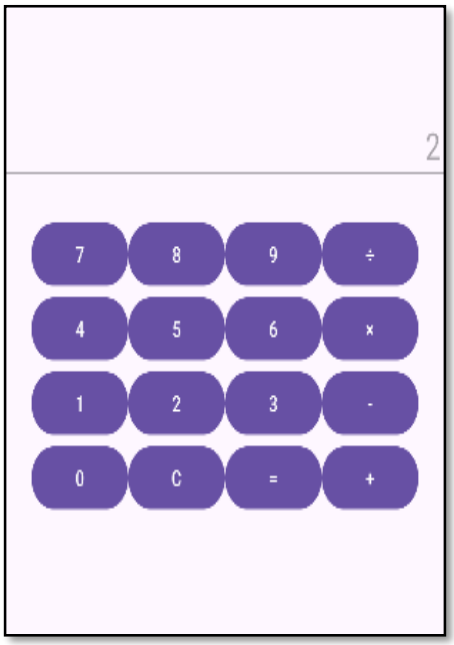
```
package com.example.calcidemo;

import android.os.Bundle;
import android.view.View;
import android.widget.Button;
import android.widget.EditText;

import androidx.activity.EdgeToEdge;
import androidx.appcompat.app.AppCompatActivity;
import androidx.core.graphics.Insets;
import androidx.core.view.ViewCompat;
import androidx.core.view.WindowInsetsCompat;
```

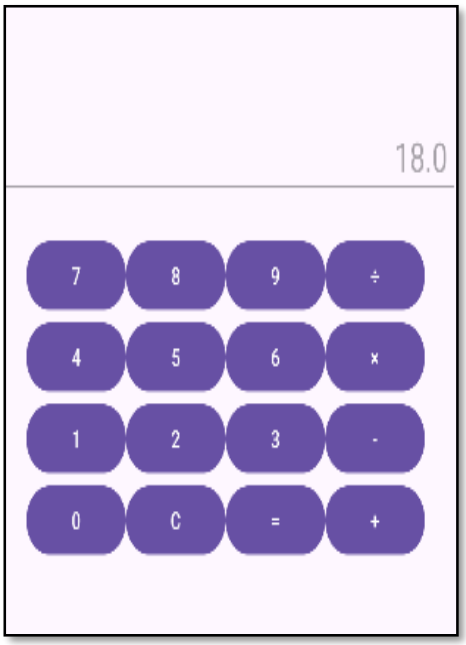
```
public class MainActivity extends AppCompatActivity {
    EditText etDisplay;
    double num1 = 0, num2 = 0;
    String operator = "";

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        EdgeToEdge.enable(this);
        setContentView(R.layout.activity_main);
        etDisplay = findViewById(R.id.etDisplay);
        ViewCompat.setOnApplyWindowInsetsListener(findViewById(R.id.main), (v, insets) -> {
            Insets systemBars = insets.getInsets(WindowInsetsCompat.Type.systemBars());
            v.setPadding(systemBars.left, systemBars.top, systemBars.right, systemBars.bottom);
            return insets;
        });
    }
    public void onNumberClick(View view)
    {
        Button button=(Button) view;
        etDisplay.append(button.getText().toString());
    }
    public void onOperatorClick(View view) {
        Button button = (Button) view;
        operator = button.getText().toString();

        if (!etDisplay.getText().toString().isEmpty()) {
            num1 = Double.parseDouble(etDisplay.getText().toString());
            etDisplay.setText("");
        }
    }
    public void onEqualClick(View view) {
        if (!etDisplay.getText().toString().isEmpty()) {
            num2 = Double.parseDouble(etDisplay.getText().toString());
            double result = 0;

            switch (operator) {
                case "+":
                    result = num1 + num2;
                    break;
                case "-":
                    result = num1 - num2;
                    break;
                case "×":
                    result = num1 * num2;
                    break;
                case "÷":
                    if (num2 != 0)
                        result = num1 / num2;
                    else {
                        etDisplay.setText("Error");
```

```
                return;
            }
            break;
        }

        etDisplay.setText(String.valueOf(result));
    }
}

// Clear button click
public void onClearClick(View view) {
    etDisplay.setText("");
    num1 = num2 = 0;
    operator = "";
}
}
```

**OUTPUT**



**EX. No. 08** **LIST VIEW**

## AIM

To develop an application that shows different country name on a list view and on selecting an item it displays its corresponding flag.

## PROCEDURE

1. Start an android studio application.
2. Select File in the menu bar.
3. Choose your project as **Empty View Activity** and click the Next button.
4. Name your project (The application name begins with an uppercase letter), select language as java, select your minimum API level as (Android 8.0 Oreo) then click Finish button.
5. For designing, type xml coding in **activity_main.xml**.

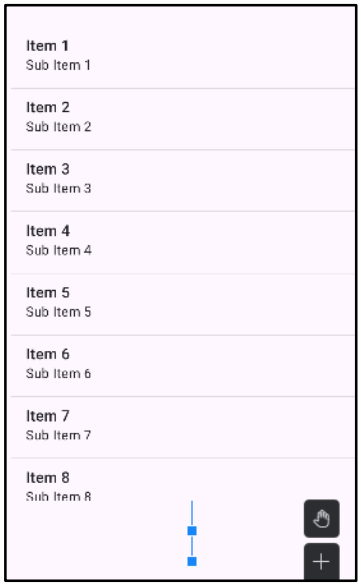
   Components used for designing

   - ListView
   - TextView
6. Type java coding in **MainActivity.java**.
7. To run the 'app' press play button in android studio or shift + F10.
8. Select Deployment project as your Emulator
9. In the connected device, select your Emulator and click the ok button.
10. Open your installed app in Emulator.

## PROGRAM

**activity_main.xml**

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:id="@+id/main"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:gravity="center"
    android:padding="16dp"
    tools:context=".MainActivity">


<LinearLayout
    android:layout_width="match_parent"
    android:layout_height="500dp"
    android:gravity="center"
    android:orientation="vertical">


<ListView
        android:layout_width="match_parent"
```



```
        android:layout_height="wrap_content"
        android:id="@+id/lv1">

</ListView>
</LinearLayout>

    <TextView
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:textSize="32sp"
        android:textStyle="bold"
        android:id="@+id/tv1"
        android:paddingTop="20dp">
    </TextView>

</LinearLayout>
```

**MainActivity.java**

```
package com.example.listview;

import android.os.Bundle;
import android.view.View;
import android.widget.AdapterView;
import android.widget.ArrayAdapter;
import android.widget.ListView;
import android.widget.TextView;

import androidx.appcompat.app.AppCompatActivity;
import androidx.core.graphics.Insets;
import androidx.core.view.ViewCompat;
import androidx.core.view.WindowInsetsCompat;

public class MainActivity extends AppCompatActivity {
    TextView tv1;
    ListView lv1;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
        tv1=findViewById(R.id.tv1);
        lv1=findViewById(R.id.lv1);
        ViewCompat.setOnApplyWindowInsetsListener(findViewById(R.id.main), (v, insets) -> {
            Insets systemBars = insets.getInsets(WindowInsetsCompat.Type.systemBars());
            v.setPadding(systemBars.left, systemBars.top, systemBars.right, systemBars.bottom);
            return insets;
        });
        String[] countries = {
                "Argentina","Australia","Brazil","Canada",
                "China","France", "Germany","India",
                "Italy","Japan", "Mexico","Netherlands",
                "New Zealand","Russia", "Singapore","South Africa",
```

```
            "Spain","Sweden", "United Kingdom","United States"
        };
        ArrayAdapter<String> adapter=new ArrayAdapter<>(
                        this, android.R.layout.simple_list_item_1,countries);
        lv1.setAdapter(adapter);

        lv1.setOnItemClickListener(new AdapterView.OnItemClickListener() {
            @Override
            public void onItemClick(AdapterView<?> parent, View view, int position, long id) {
                tv1.setText("Selected Country: " + countries[position]);
            }
        });


    }
}
```

**OUTPUT**